

Microeconomics

Lecture 14: Deadweight Loss and Administered Prices

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What We Know

Tools for measuring welfare:

- Consumer surplus: $CS = \int_0^{Q^*} [P_D(Q) - P^*] dQ$ —for linear demand, $\frac{1}{2} Q^* (P_{\max}^D - P^*)$
- Producer surplus: $PS = \int_0^{Q^*} [P^* - P_S(Q)] dQ$ —for linear supply with zero intercept, $\frac{1}{2} Q^* P^*$
- Total surplus: $TS = CS + PS$
- **First Welfare Theorem:** the competitive equilibrium maximizes TS

Now we measure the distortion

Any deviation from the competitive equilibrium (quantity $\tilde{Q} \neq Q^*$, price $\tilde{P} \neq P^*$) reduces TS. The reduction is called **deadweight loss** (DWL). We learn how to compute it, then apply it to the **maximum price** and the **minimum price** imposed by the government.

Lecture Objectives

1. **Define** deadweight loss as a difference in welfare
2. **Compute** the DWL with the triangle formula $\frac{1}{2}(Q^* - \tilde{Q})(P_D(\tilde{Q}) - P_S(\tilde{Q}))$
3. **Analyze** the mechanism of a *price ceiling* (maximum price) below equilibrium
4. **Compute** ΔCS , ΔPS and DWL with a *price ceiling*
5. **Analyze** the mechanism of a *price floor* (minimum price) above equilibrium
6. **Compute** ΔCS , ΔPS and DWL with a *price floor*
7. **Discuss** who gains and who loses from each intervention
8. **Solve** complete numerical exercises

Reference market we will use

$Q^D = 80 - 2P$ (inv. $P_D = 40 - Q/2$) and $Q^S = 3P$ (inv. $P_S = Q/3$).

Equilibrium: $P^* = 16$, $Q^* = 48$, $CS = 576$, $PS = 384$, $TS = 960$.

Why Intervene in the Market?

The First Welfare Theorem states that the competitive market, under ideal conditions, maximizes TS. If so, why do governments intervene so often?

Three underlying reasons:

1. **Market failures.** The “ideal” equilibrium requires the absence of market power, externalities, public goods, information asymmetries. When these are present the First Welfare Theorem does not hold and intervention can improve efficiency.
2. **Distributional considerations.** The First Welfare Theorem guarantees efficiency but not equity. The equilibrium may be socially unacceptable: rents unaffordable for low incomes, wages squeezed too low, access to medicines blocked by price.
3. **Government failures.** Paradoxically, interventions often worsen the situation they intend to correct. It is essential to evaluate their effects rigorously before deciding on them.

The economist's task

Not to prescribe whether to intervene (a political choice), but to measure the costs and benefits of each intervention.

Deadweight Loss—Definition

Deadweight Loss (DWL)

Given a market with equilibrium (P^*, Q^*) , the deadweight loss of a traded quantity $\tilde{Q} \neq Q^*$ is the fall in total surplus:

$$DWL = TS(Q^*) - TS(\tilde{Q}) \geq 0$$

Intuition. At Q^* all efficient trades (those with $P_D \geq P_S$) are realized. If the market trades less than Q^* :

- For every unit between $\tilde{Q}$ and Q^* : $P_D(Q) > P_S(Q)$
- Those units were worth trading (value to the consumer $>$ cost to the producer)
- Since they are not produced, that potential gain from trade **vanishes**

$DWL \neq \text{transfer}$

The DWL is not money passing from one party to another. It is value that **disappears**: it goes to neither consumers, producers, nor the government. That is why it is the pure measure of inefficiency.

The Triangle Formula

With linear curves, if the traded quantity falls from Q^* to $\tilde{Q} < Q^*$:

$$DWL = \frac{1}{2} \cdot \underbrace{(Q^* - \tilde{Q})}_{\text{base}} \cdot \underbrace{(P_D(\tilde{Q}) - P_S(\tilde{Q}))}_{\text{height}}$$

Reading the two factors:

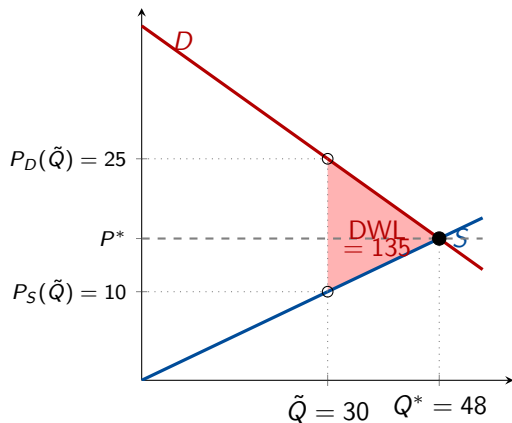
- **Base:** how far the traded quantity has moved from the optimum
- **Height:** how much the “lost” unit was worth (P_D) relative to what it cost to produce (P_S), at the last unit actually traded $\tilde{Q}$

Alternative formula

$$DWL = TS(Q^*) - \int_0^{\tilde{Q}} [P_D(Q) - P_S(Q)] dQ$$

Useful when the curves are not linear.

DWL: Graph and Computation



Scenario: for some reason the traded quantity is $\tilde{Q} = 30$ instead of $Q^* = 48$.

$$P_D(30) = 40 - 15 = 25$$

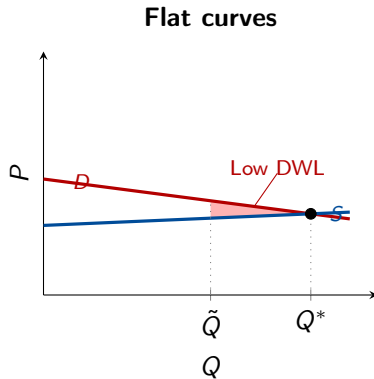
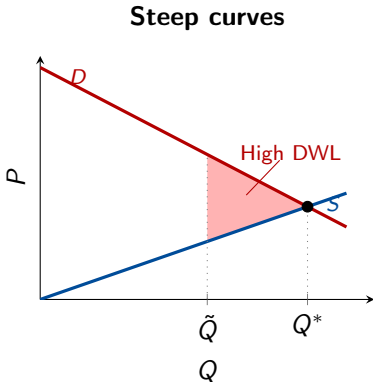
$$P_S(30) = 30/3 = 10$$

$$\begin{aligned} DWL &= \frac{1}{2} \cdot (48 - 30) \cdot (25 - 10) \\ &= \frac{1}{2} \cdot 18 \cdot 15 = 135 \end{aligned}$$

Interpretation: the 18 units not produced were worth on average $\frac{25+16}{2} = 20.5$ to consumers and could be produced on average at $\frac{10+16}{2} = 13$. On average, each would generate a gain from trade of $\frac{25-10}{2} = 7.5$. Multiplying by 18 units: 135.

DWL and Elasticity: Visual Comparison

For the same reduction in quantity $Q^* - \tilde{Q}$, the DWL depends on the slopes of D and S .



Intuition. The height of the DWL triangle is $P_D(\tilde{Q}) - P_S(\tilde{Q})$: with steep curves it grows quickly, with flat curves slowly. Policy implication: for the same contraction in quantity, interventions in more elastic markets produce less DWL.

DWL: General Formula

The triangle formula holds exactly only for linear D and S . In the general case:

General formulation with integrals

$$\text{TS}(Q) = \int_0^Q [P_D(q) - P_S(q)] dq$$

Total surplus at quantity Q is the integral of the “marginal gain from trade” over all units traded. TS is maximized at $Q = Q^*$, where $P_D(Q^*) = P_S(Q^*)$.

$$\text{DWL}(\tilde{Q}) = \text{TS}(Q^*) - \text{TS}(\tilde{Q}) = \int_{\tilde{Q}}^{Q^*} [P_D(q) - P_S(q)] dq$$

DWL: Checking the Linear Case

With $P_D = 40 - Q/2$, $P_S = Q/3$:

$$\int_{30}^{48} \left[40 - \frac{5q}{6} \right] dq = \left[40q - \frac{5q^2}{12} \right]_{30}^{48} = 960 - 825 = 135 \checkmark$$

This coincides with the triangle formula $\frac{1}{2} \cdot 18 \cdot 15 = 135$. With linear curves the two formulas are equivalent; with nonlinear curves only the integral gives the correct value.

Price Ceiling—Definition

Price ceiling (maximum price)

The government imposes by law a maximum price $\bar{P} < P^*$: no seller may charge a price above $\bar{P}$.

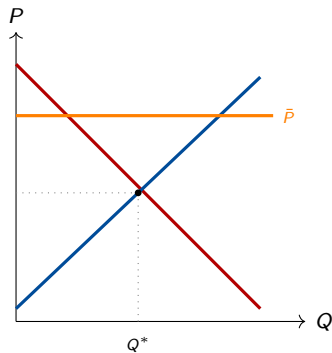
Why? The stated goal is to protect consumers. Examples:

- Rent control: Stockholm, New York, Berlin
- Prices of essential medicines: many countries regulate the price of life-saving drugs
- Price of bread/wheat in times of war or famine
- Regulated electricity tariffs for household use

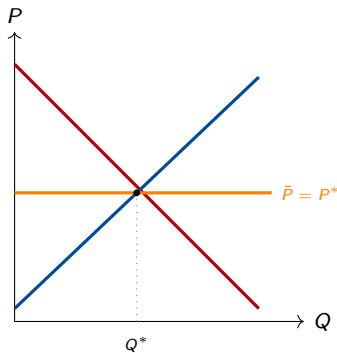
A maximum price is binding only if $\bar{P} < P^*$

If $\bar{P} \geq P^*$, the market would still be free to operate at $P^* < \bar{P}$: the ceiling does not bite. The interesting analytical consequences arise when $\bar{P} < P^*$.

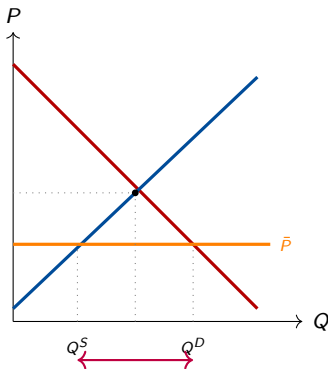
Price Ceiling: The Three Cases



Case 1: $\bar{P} > P^*$
not binding



Case 2: $\bar{P} = P^*$
borderline



Case 3: $\bar{P} < P^*$
binding (excess D)

Only the third case is analytically interesting. The market contracts to $Q^S(\bar{P}) < Q^*$, excess demand $Q^D(\bar{P}) - Q^S(\bar{P}) > 0$ arises, and CS, PS and DWL differ from equilibrium.

Price Ceiling: Mechanism and Excess Demand

With $\bar{P} = 10$ in our market ($P^* = 16$):

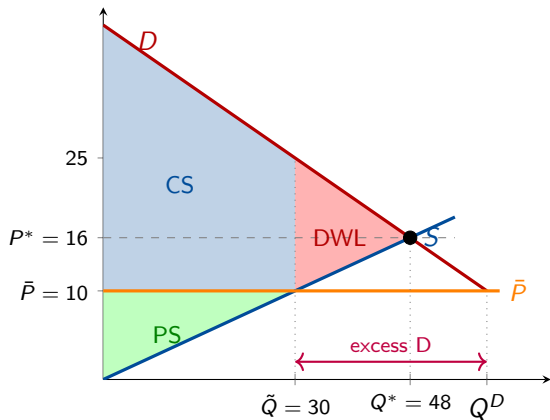
- At $P = 10$: $Q^D = 80 - 20 = 60$, $Q^S = 30$
- Consumers would like to buy 60 units
- Producers are willing to sell only 30 units (at that price, MC beyond that is not covered)
- **Short side of the market:** $\tilde{Q} = \min(Q^D, Q^S) = 30$
- **Excess demand:** $60 - 30 = 30$ units

The traded quantity is determined by supply

For $\bar{P} < P^*$ the binding constraint is supply: at that price producers do not find it profitable to produce more. The market stops at $\tilde{Q} = Q^S(\bar{P})$.

Allocation problem. 30 available units must be shared among 60 potential buyers: a rationing problem arises (queues, waiting lists, preference for existing customers, black market).

Price Ceiling: Graph of the Areas



Components at price $\bar{P} = 10$, $\tilde{Q} = 30$:

CS (trapezoid): area between D and $\bar{P}$ from 0 to $\tilde{Q}$.

$$CS = \int_0^{30} (40 - Q/2 - 10) dQ$$

$$= \left[30Q - \frac{Q^2}{4} \right]_0^{30} = 900 - 225 = 675$$

PS (triangle): area between $\bar{P}$ and S from 0 to $\tilde{Q}$.

$$PS = \frac{1}{2} \cdot 30 \cdot 10 = 150$$

DWL: triangle between D , S from $\tilde{Q}$ to Q^* .

$$DWL = \frac{1}{2} \cdot 18 \cdot 15 = 135$$

Price Ceiling: Welfare Changes

	Equilibrium ($P^* = 16$)	Ceiling ($\bar{P} = 10$)	Δ
CS	576	675	+99
PS	384	150	-234
TS	960	825	-135
DWL	0	135	

Does CS rise or fall?

Aggregate CS **rises** (+99), but only if we assume efficient rationing (the 30 units go to those with the highest WTP). Those who buy get the unit at $\bar{P} = 10$ instead of $P^* = 16$, saving 6 per unit $\times$ 30 units = 180. But they lose the CS on the units between 30 and 48 that would have been bought at $P^* = 16$.

Who loses:

- Producers (they lose 234: lower quantity *and* lower price)
- Rationed consumers, i.e., the 18 who would have bought at P^* but now find no one willing to sell to them

Does CS Really Increase with a Ceiling?

The standard analysis gives $\Delta CS = +99$ with $\bar{P} = 10$. But this conclusion rests on a **crucial hidden premise**: the 30 available units go to those who value them most.

Two rationing scenarios:

1. **Efficient rationing** (model assumption): the 30 units go to the consumers with the highest WTP (those with $q \in [0, 30]$, i.e., WTP between 25 and 40 €/unit). The average WTP is $\frac{40+25}{2} = 32.5$.
 $\Rightarrow CS = 30 \cdot (32.5 - 10) = 675$
2. **Random rationing** (queues, lottery, “first-come first-served”): the 30 units go to 30 consumers drawn uniformly from the 60 who would like to buy. The WTPs of the 60 range from 10 to 40, so the average WTP of a random buyer is $\frac{40+10}{2} = 25$.
 $\Rightarrow CS = 30 \cdot (25 - 10) = 450$

Does CS Really Increase with a Ceiling? (2/2)

To summarize:

	CS	Δ CS vs equilibrium
Competitive equilibrium	576	—
Ceiling, efficient rationing	675	+99
Ceiling, random rationing	450	-126

Result

The consumers' gain from the ceiling is **entirely hypothetical**: it exists only if rationing is optimal. With random allocation (typical in reality: queues, waiting lists), CS can *collapse* below the equilibrium level. With random rationing the DWL does not stay at 135: PS is unchanged (150), but on top of the loss from lower quantity comes the loss from misallocation ($675 - 450 = 225$), for a total DWL of 360.

Price Ceiling: The Rationing Problem

With excess demand, the 30 available units must be allocated among 60 potential buyers. The market *by construction* cannot do this through the price: who decides who buys?

Observed rationing mechanisms:

- **First-come, first-served:** queues, waiting lists, limited opening hours. Rewards those with “time” available (not those with the highest WTP).
- **Lottery / random draw:** random allocation. Fair *ex ante* (everyone has the same probability) but inefficient *ex post* (the good may go to someone who does not value it).
- **Seller discretion:** the producer chooses (friends, relatives, loyal customers). Opens the door to discrimination and corruption.
- **Black market:** illegal transactions at prices above the maximum. Restores market prices but adds legal risk.
- **Administrative rationing:** ration cards and coupons issued by the government in fixed quantities.

Efficient allocation $\neq$ automatic

In computing CS in our example (+99) we implicitly assumed that the 30 units go to those with the highest WTP. In practice, rationing by queues or lottery leads to a worse allocation: actual CS may be *lower* than 675—and potentially *below* equilibrium CS.

Price Ceiling: Quality Effects and Black Market

The fixed-curve model underestimates the distortions. Producers who cannot use price as a lever respond with other tools.

Quality reduction. If $\bar{P} < P^*$ the producer can, instead of reducing quantity, keep Q^* but offer a lower-quality good. The “implicit price” per unit of quality returns to the market level.

Evidence. Rent-controlled apartments see less maintenance and renovation (Olsen, JPE 1972; Sims, 2007). Quality becomes the unregulated adjustment variable.

Black market. If the gap $P^* - \bar{P}$ is large, illegal transactions emerge at the “hidden” equilibrium price. The model’s apparent DWL does not materialize—CS and PS return to their original values—but legal, search and trust costs still reduce welfare.

Reversal of bargaining power.

- Producers choose their customers (“bonuses” from those who pay off the books)
- Consumers must prove themselves “deserving” (references, undeclared deposits)
- Rent-seeking positions (lease takeovers, subletting) become central

Price Ceiling: The Case of Rent Control

Rent control: empirical evidence

- **Stockholm**: average wait for a rent-controlled municipal apartment: 9–20 years depending on the neighborhood. Long-run supply heavily compressed.
- **San Francisco**: Diamond, McQuade and Qian (AER 2019) find that rent control reduces rental supply by 15% and raises unregulated rents by 5% in the same area.
- **Berlin**: the *Mietendeckel* (2020) reduced rents on covered units by 7–11% but cut the supply of new listings by about 30%. Struck down by the Constitutional Court in 2021.

Two empirical lessons:

1. Long-run supply is **much more elastic** than short-run supply: the DWL grows over time.
2. The distributional effects are ambiguous: the beneficiaries are the *incumbents* (those who already have a regulated lease), not new entrants to the labor market or immigrants.

Price Ceiling: Extreme Historical Cases

World War II: OPA rationing (USA)

The *Office of Price Administration* (1941–1946) imposed ceilings on about 90% of consumer goods, to prevent shortage-driven inflation and free up resources for the war effort.

- **Ration cards** for sugar, coffee, meat, gasoline, tires
- Significant **black market**: estimates of 7–15% of actual GDP
- After the war, prices rose by 20% within a few months (backlog of supply and pent-up demand released all at once)

1973 oil crisis (USA)

After the OPEC embargo, the gasoline price ceilings introduced by Nixon in 1971 remained in force (average price $\approx$ 0.40 USD/gal in 1973). Result: *miles-long queues* (waits of several hours) and odd-even license plate rationing. Frank (2015) estimates that the cost of time lost in queues exceeded the value of the price savings.

Price Floor: Definition and Motivation

Price floor (minimum price)

The government imposes by law a minimum price $\underline{P} > P^*$: no transaction may take place at a price below $\underline{P}$.

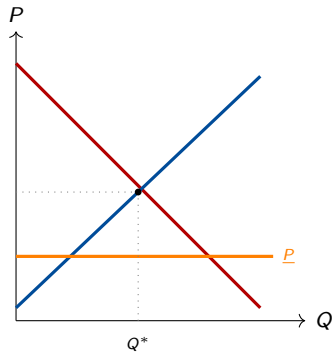
Why? The stated goal is to protect sellers. Examples:

- **Minimum wage** ($\underline{P}$ in the labor market): USA, EU, Italy (ongoing debate)
- **Agricultural minimum prices**: the EU CAP (Common Agricultural Policy), minimum prices for milk, sugar, cereals
- Minimum fees for professional services (debate on liberalization)

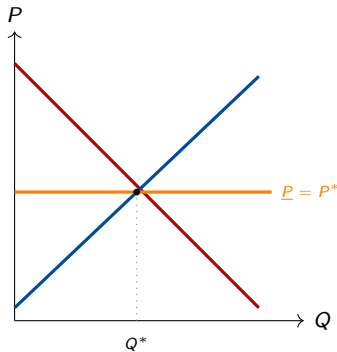
A minimum price is binding only if $\underline{P} > P^*$

If $\underline{P} \leq P^*$, the market would operate at $P^* \geq \underline{P}$: the floor does not bite. The analytical consequences emerge when $\underline{P} > P^*$.

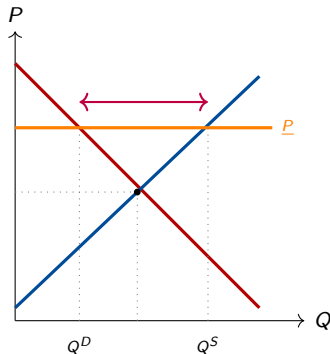
Price Floor: The Three Cases



Case 1: $\underline{P} < P^*$
not binding



Case 2: $\underline{P} = P^*$
borderline



Case 3: $\underline{P} > P^*$
binding (excess S)

Only the third case has real effects. The market contracts to $Q^D(\underline{P}) < Q^*$, and excess supply $Q^S(\underline{P}) - Q^D(\underline{P}) > 0$ emerges. In the labor market: involuntary unemployment.

Price Floor: Mechanism and Excess Supply

With $\underline{P} = 25$ in our market ($P^* = 16$):

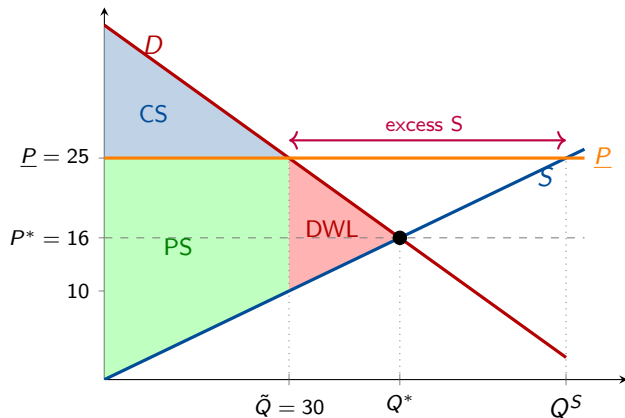
- At $P = 25$: $Q^D = 80 - 50 = 30$, $Q^S = 75$
- Producers would like to sell 75 units
- Consumers are willing to buy only 30
- **Short side:** $\tilde{Q} = \min(Q^D, Q^S) = 30$
- **Excess supply:** $75 - 30 = 45$ units

The traded quantity is determined by demand

For $\underline{P} > P^*$ the binding constraint is demand: at that price consumers do not buy more. The market stops at $\tilde{Q} = Q^D(\underline{P})$.

The unsold surplus problem. 45 units remain unsold. In the labor market: involuntary unemployment. In agricultural products: accumulated surpluses (the CAP's historic “butter mountains”) often withdrawn by the government.

Price Floor: Graph of the Areas



Components at price $\underline{P} = 25$, $\tilde{Q} = 30$:

CS (triangle): area between D and $\underline{P}$ from 0 to $\tilde{Q}$.

$$CS = \frac{1}{2} \cdot 30 \cdot (40 - 25) = 225$$

PS (trapezoid): area between $\underline{P}$ and S from 0 to $\tilde{Q}$.

$$\begin{aligned} PS &= \int_0^{30} (25 - Q/3) dQ \\ &= \left[25Q - \frac{Q^2}{6} \right]_0^{30} = 750 - 150 = 600 \end{aligned}$$

DWL: triangle between D , S from $\tilde{Q}$ to Q^* .

$$DWL = \frac{1}{2} \cdot 18 \cdot 15 = 135$$

Price Floor: Welfare Changes

	Equilibrium ($P^* = 16$)	Floor ($\underline{P} = 25$)	Δ
CS	576	225	-351
PS	384	600	+216
TS	960	825	-135
DWL	0	135	

A notable symmetry

The price ceiling ($\bar{P} = 10$) and the price floor ($\underline{P} = 25$) yield the **same** $\tilde{Q} = 30$ and the **same** DWL of 135, because the welfare loss depends only on the deviation of quantity from equilibrium, not on which side of the market is rationed. What changes is the **distribution**: the ceiling hurts producers, the floor hurts consumers.

Who gains: producers who manage to sell (216 more). **Who loses:** consumers (all of them; those who buy pay more) and producers who find no buyers (excess of 45 units).

Price Floor: The Minimum Wage

Minimum wage: the empirical debate

- **Card & Krueger (AER 1994)**: classic NJ vs PA study; moderate minimum wage increases do not seem to reduce employment in the fast-food sector
- **Cengiz et al. (QJE 2019)**: using 138 state-level changes, they estimate that moderate minimum wages (up to 59% of the median) do not reduce aggregate employment
- **Neumark & Wascher**: negative effects on low-skilled workers, especially the young

Why can the standard model fail? Three main reasons:

1. **Monopsony**: an employer with market power already hires less than L^* ; a moderate minimum wage can *increase* employment (see Course Notes, Ch. 12)
2. **Income effect**: higher wages can boost aggregate demand, partially offsetting unemployment
3. **Pass-through to prices**: increases may be passed on to final prices rather than reducing employment

Price Floor: The EU Common Agricultural Policy

The CAP is the historic European system of farm income support introduced by the Treaty of Rome (1957) and consolidated from the 1960s.

Guaranteed-price mechanism (until the 1992 MacSharry reform):

1. The Community set a **guaranteed minimum price** $\underline{P}$ for each product (cereals, milk, sugar, beef. . .)
2. If the market price fell below $\underline{P}$, the EU bought all unsold output at the minimum price
3. Import tariffs prevented world prices from driving down the domestic price

Consequence: the famous “surpluses”. “Butter mountains” (over 1 Mt stored in the 1980s), “wine lakes” (up to 3 billion liters), and millions of tonnes of wheat, poultry and sugar withdrawn from the market. Costs: storage, spoilage, resale below cost to third countries (dumping challenged in the GATT/WTO), amounting to €10–15 billion a year.

From 1992 (MacSharry), guaranteed prices were cut and offset by per-hectare payments still tied to crops; from 2003 (Fischler), **decoupled direct payments**: farmers are paid per *hectare* regardless of what they produce—a lump-sum equivalent (less distortionary), but politically more entrenched.

Price Floor: Long-Run Adjustment

The static model underestimates the long-run effects of a price floor, especially in the labor market.

In the short run (labor demand and supply fairly inelastic):

- Employment falls relatively little
- The floor is mainly a transfer from employers to workers who remain employed
- The apparent DWL is modest

In the long run (labor demand and supply more elastic: choices about education, migration, participation):

- Low-skilled labor supply contracts (those who find work are discouraged, those who do not change sector)
- Employers invest in **automation**: they substitute capital for labor
- With a binding floor, employment is $L^D(\underline{w})$: more elastic demand reduces it; a supply contraction reduces unemployment, not employment
- The DWL grows

Two implications

1. Short-run cross-sectional studies (a single state, a few years) systematically underestimate the cost of the floor. Long-run studies (Neumark & Wascher) tend to estimate larger costs.
2. The same logic applies to rent control: the supply of apartments contracts only gradually (existing owners have little room to adjust in the SR; in the LR no new regulated apartments are built).

Energy Price Caps: The 2022–2023 Case

A large-scale natural experiment in price ceilings.

Context. Russian invasion of Ukraine (Feb. 2022) $\Rightarrow$ European gas prices $\times 10$ within a few months. The wholesale price reached > 300 €/MWh in August 2022 (versus 20–25 €/MWh before the crisis).

Responses of European governments (de facto price ceilings):

- **Italy:** “Mercato Tutelato” (protected market)—prices frozen for vulnerable households; cost over €60 billion in 2022–23
- **France:** *bouclier tarifaire*—increases capped at +4% for households; cost $\approx$ €45 billion
- **Germany:** *Gaspreisbremse*—maximum price of 12 cents/kWh up to 80% of baseline consumption; cost $\approx$ €200 billion

Reading it through the model of this lecture:

- CS protected by the cap (relative to the free market with prices $\times 10$)
- PS of suppliers (Eni, Enel, etc.) limited, but offset by government subsidies
- Fiscal cost: revenue from taxing “windfall profits” + public debt
- DWL: efficient energy-saving consumption under-incentivized; weaker price signals for renewables

When the ceiling is justified

Useful if the shock is temporary and the market cannot readjust quickly. In the LR, it blocks the incentives for energy efficiency and renewables—precisely what would be needed to prevent the next crisis.

Ceiling vs Floor: Full Comparison

	Equilibrium	Ceiling $\bar{P} = 10$	Floor $\underline{P} = 25$
Quantity traded	$Q^* = 48$	$\tilde{Q} = 30$	$\tilde{Q} = 30$
Imbalance	–	Excess D = 30	Excess S = 45
Constrained side	–	Supply	Demand
Price paid	16	10	25
CS	576	675 (+99)	225 (–351)
PS	384	150 (–234)	600 (+216)
TS	960	825	825
DWL	0	135	135

Insights:

- For the same distorted quantity ($\tilde{Q} = 30$), the two interventions produce *the same* DWL: inefficiency depends only on the distance from Q^* .
- The distributional effects are *mirror images*: the ceiling redistributes from producers to the consumers who manage to buy, the floor does the opposite. In both cases part of these transfers reaches no one (the DWL).

Effects Not Captured: Price Ceiling

The model with fixed curves and a single price underestimates real distortions. With a price ceiling, systematic effects emerge that the triangle analysis does not capture.

- **Quality reduction:** if the price cannot rise, the producer cuts quality (maintenance of rent-controlled apartments, freshness of food, delivery times)
- **Queues and search time:** “first-come” rationing has a cost in hours lost. Frank (2015) estimates that during the '73 crisis the value of time lost in queues exceeded the price savings
- **Black market:** illegal transactions at prices above the legal maximum, with search costs and legal risk
- **Inefficient allocation:** whoever obtains the good is not necessarily whoever values it most. Random rationing lowers CS further
- **Long-run investment:** LR supply contracts more than SR supply (investment in new supply does not happen). Diamond-McQuade-Qian (2019) estimate a 15% reduction in the supply of regulated rentals in San Francisco

Implication for policy analysis. The triangle DWL is a *lower bound* on the social cost of a price ceiling. Real costs are systematically larger, especially in the LR.

Effects Not Captured: Price Floor

Similarly, with a price floor distortions emerge that the basic model does not account for.

- **Waste of the unsold surplus:** in the CAP case, surpluses worth billions of € accumulated, often destined for destruction or resale below cost
- **Greater selectivity:** in the labor market, the employer chooses only the most productive candidates, excluding the less skilled (amplified “screening” effect)
- **Substitution investment:** employers automate (substitute capital for labor) when the cost of labor is artificially high. This reduces LR labor demand
- **Administrative costs:** monitoring compliance, managing surpluses, enforcing the minimum
- **Internal distribution:** between those who “manage to sell” at the floor and those left out. Small producers (or low-skilled workers) are often the first to be excluded

A common effect

With both a ceiling and a floor: the intervention generates a **double layer of distributional effects**. The first is visible in the model (CS vs PS). The second is between “insiders” (those who manage to trade at the administered price) and “outsiders” (those excluded), and is often invisible in aggregate statistics.

When Is a Price Control Justifiable?

The competitive model leads to a clear-cut conclusion: administered prices reduce total surplus. Yet the empirical evidence shows that many countries use them anyway. When can intervention be rational?

1. **Non-competitive market.** If the seller has market power (monopoly, monopsony, cartel), the market price is not P^* but higher (or lower under monopsony). A moderate ceiling can *increase* efficiency by bringing the price back toward marginal cost. *We will see this formally in L16 (monopoly).*
2. **Externalities or merit goods.** If the good generates social benefits undervalued by the market (vaccines, education, decent housing), the market price under-incentivizes consumption. A subsidy can correct this—but it must be sized to the externality, not set arbitrarily.
3. **Distributional constraints.** If society deems the market allocation unacceptable (no access to essential goods for the poor, wages below the subsistence threshold), it knowingly accepts the efficiency loss in exchange for equity. The DWL becomes the “moral cost” of the intervention.
4. **Temporary shocks.** During crises (war, pandemic, sudden scarcity), the market price would rise to politically unsustainable levels. A temporary ceiling accompanied by explicit rationing may be the lesser evil.

The role of economic analysis

DWL analysis does not say “never intervene”. It says: “measure the cost of the intervention and compare it with the benefit (correcting a failure, equity, social stability)”. Intervention is justified when the benefit exceeds the DWL—a choice that remains political.

Exercise 1—Price Ceiling

Market with $Q^D = 80 - 2P$, $Q^S = 3P$. Equilibrium: $P^* = 16$, $Q^* = 48$, $TS = 960$.
The government imposes $\bar{P} = 12$.

- (a) Determine the quantity demanded, supplied and traded. Is there excess demand or excess supply?
- (b) Compute $P_D(\tilde{Q})$ and $P_S(\tilde{Q})$.
- (c) Compute CS, PS and DWL.
- (d) Compare with the equilibrium: who gains, who loses?

Exercise 1—Solution

(a) $Q^D(12) = 80 - 24 = 56$, $Q^S(12) = 36$. $\tilde{Q} = \min(56, 36) = 36$.

Excess demand $= 56 - 36 = 20$ units.

(b) $P_D(36) = 40 - 18 = 22$; $P_S(36) = 36/3 = 12$.

(c)

$$CS = \int_0^{36} (40 - Q/2 - 12)dQ = \left[28Q - \frac{Q^2}{4} \right]_0^{36} = 1,008 - 324 = 684$$

$$PS = \frac{1}{2} \cdot 36 \cdot 12 = 216$$

$$DWL = \frac{1}{2}(48 - 36)(22 - 12) = \frac{1}{2} \cdot 12 \cdot 10 = 60$$

Check: $TS = 684 + 216 = 900 = 960 - 60 \checkmark$

(d) $\Delta CS = +108$ (consumers who buy), $\Delta PS = -168$ (producers). $\Delta TS = -60$ (DWL). They produce less and at a lower price: producers bear the largest loss.

Exercise 2—Price Floor

Same market. The government imposes $\underline{P} = 22$.

- (a) Determine the quantity demanded, supplied and traded. Is there excess demand or excess supply?
- (b) Compute $P_D(\tilde{Q})$ and $P_S(\tilde{Q})$.
- (c) Compute CS, PS and DWL.
- (d) Compare the DWL with Exercise 1. Why is it the same?

Exercise 2—Solution

(a) $Q^D(22) = 80 - 44 = 36$, $Q^S(22) = 66$. $\tilde{Q} = \min(36, 66) = 36$.

Excess supply = $66 - 36 = 30$ units.

(b) $P_D(36) = 22$ (it coincides with the floor: demand binds at that price); $P_S(36) = 12$.

(c)

$$CS = \frac{1}{2} \cdot 36 \cdot (40 - 22) = 324$$

$$PS = \int_0^{36} (22 - Q/3) dQ = \left[22Q - \frac{Q^2}{6} \right]_0^{36} = 792 - 216 = 576$$

$$DWL = \frac{1}{2}(48 - 36)(22 - 12) = 60$$

Check: $TS = 324 + 576 = 900 = 960 - 60$ ✓

(d) The DWL is identical to Ex. 1 because the traded quantity $\tilde{Q} = 36$ is the same, as is the gap between $P_D(\tilde{Q})$ and $P_S(\tilde{Q})$. Only the redistribution changes: CS and PS “swap” relative to Ex. 1 (producers gain 192, consumers lose 252).

Exercise 3—True or False (Part I)

Evaluate each statement (True/False) and briefly explain.

(a) “A price ceiling always helps all consumers.”

(b) “A price floor in the labor market always reduces unemployment.”

(a) False. It helps only those who manage to buy. Those who are rationed lose the CS they had at equilibrium. Aggregate CS can even fall if rationing is random and does not go to those with the highest WTP. Moreover, reduced quality and the black market further erode the benefits.

(b) False—in fact, it typically creates unemployment. A minimum wage $\underline{w} > w^*$ in the labor market generates excess supply (job seekers) relative to demand (jobs offered by firms). Exception: a monopsony market (see Course Notes, Ch. 12), where a moderate minimum wage can *increase* employment.

Exercise 3—True or False (Part II)

- (c) “If the DWL is zero, there has been no distributional effect.”
- (d) “A price ceiling at a level $\bar{P} > P^*$ has no effect on the market.”

(c) False. DWL and distribution are different things. At the competitive equilibrium the DWL is zero, but CS and PS are both well defined and nonzero—there is a precise distribution of welfare, there is simply no net efficiency loss. A lump-sum transfer from producer to consumer would have $DWL = 0$ while still redistributing welfare.

(d) True. If $\bar{P} \geq P^*$, the market would operate at $P^* \leq \bar{P}$: the ceiling is not binding, no transaction is affected by the constraint, and the equilibrium is identical to the case without intervention. Symmetrically for a floor with $\underline{P} \leq P^*$.

Exercise 4—Minimum Wage

Low-skilled labor market: demand $L^D = 600 - 10w$, supply $L^S = 20w$ (where w is the hourly wage in €).

- (a) Find w^* and L^* .
- (b) The government introduces a minimum wage $\underline{w} = 24$. How many workers are employed? How many are unemployed?
- (c) Compute workers' surplus, firms' surplus and DWL in the new situation.
- (d) Does the minimum wage improve aggregate workers' welfare?

Exercise 4—Solution

(a) $600 - 10w = 20w \Rightarrow 30w = 600 \Rightarrow \boxed{w^* = 20, L^* = 400}$.

Inverse demand: $w_D(L) = (600 - L)/10$, with intercept 60. Inverse supply: $w_S(L) = L/20$.

(b) At $\underline{w} = 24$: $L^D(24) = 360$, $L^S(24) = 480$. Firms hire $\tilde{L} = 360$.

Involuntary unemployment = $480 - 360 = 120$ workers.

(c) Firms' surplus (area between labor demand and $\underline{w}$):

$$FS = \frac{1}{2} \cdot 360 \cdot (60 - 24) = 6480$$

Surplus of employed workers (area between $\underline{w}$ and supply):

$$WS = \int_0^{360} (24 - L/20) dL = \left[24L - \frac{L^2}{40} \right]_0^{360} = 8,640 - 3,240 = 5400$$

$$DWL = \frac{1}{2}(400 - 360)(24 - 18) = \frac{1}{2} \cdot 40 \cdot 6 = 120$$

(d) At equilibrium: $WS_{eq} = \frac{1}{2} \cdot 400 \cdot 20 = 4,000$. With the minimum wage: $WS = 5,400$, a change of +1,400. **Workers gain in aggregate**, but note: only the 360 employed gain (higher wage). The 40 who would have worked at $w = 20$ and are now left without a job lose everything: a redistribution between insiders and outsiders.

General Summary

Deadweight loss

- $DWL = TS(Q^*) - TS(\tilde{Q}) \geq 0$
- Triangle formula: $\frac{1}{2}(Q^* - \tilde{Q})(P_D(\tilde{Q}) - P_S(\tilde{Q}))$
- Net value lost: not transferred, it vanishes

Price ceiling ($\bar{P} < P^*$)

- $\tilde{Q} = Q^S(\bar{P})$; excess demand = $Q^D(\bar{P}) - Q^S(\bar{P})$
- Those who buy gain, those who are rationed lose. Producers always lose
- Real-world problems: quality, queues, black market, contracted LR supply

Price floor ($\underline{P} > P^*$)

- $\tilde{Q} = Q^D(\underline{P})$; excess supply = $Q^S(\underline{P}) - Q^D(\underline{P})$
- Those who sell gain, those who find no buyers lose. Consumers always lose
- Main application: minimum wage (with weak empirical evidence of unemployment)

Skills Acquired

- ✓ Define and compute deadweight loss (DWL) with the triangle formula
- ✓ Analyze the mechanism of a *price ceiling* and identify the short side of the market
- ✓ Compute CS, PS and DWL under a maximum price
- ✓ Analyze the mechanism of a *price floor* and the resulting excess supply
- ✓ Compute CS, PS and DWL under a minimum price
- ✓ Identify the distributional effects (winners and losers) of each intervention
- ✓ Recognize the limits of the model: quality, rationing, black market

Optional Homework Exercises

1. $Q^D = 100 - 4P$, $Q^S = 6P$. Find P^* , Q^* , CS, PS, TS.
2. Same market. The government imposes $\bar{P} = 6$. Compute $\tilde{Q}$, CS, PS and DWL. How large is the excess demand?
3. Same market. The government imposes $\underline{P} = 12$. Compute $\tilde{Q}$, CS, PS and DWL. How large is the excess supply?
4. Does a price ceiling cause a larger DWL in a market with very elastic or very inelastic supply? Explain with a graphical argument.
5. Suppose labor demand is $L^D = 1,000 - 20w$ and supply $L^S = 10w$. A minimum wage $\underline{w} = 40$ is introduced. How many people are employed? How many are unemployed? How large is the DWL?
6. Explain why the DWL of administered prices tends to be larger in the long run than in the short run (two reasons).